

StainPMS: Self-Bootstrapped Prompt-Mask Supervision for Nuclei Instance Segmentation

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Abstract—Nuclei instance segmentation in hematoxylin-and-eosin (H&E) histopathology is essential for analyzing the tissue microenvironment in computational pathology. Although recent methods built on the Segment Anything Model (SAM) couple automatic point prompts with prompt-conditioned mask decoding and reach strong accuracy, their performance is still bottlenecked by the limited precision of the auto-prompt pathway, where some nuclei are never prompted and difficult prompted nuclei stay imperfectly aligned with the ground truth. StainPMS is introduced as a self-bootstrapped stain-guided Prompt-Mask Supervision (PMS) framework that strengthens this pathway without changing the inference path or adding any learnable parameters. Specifically, during training the current model periodically infers an online coverage map on the training split and accumulates it monotonically. Residual hematoxylin peaks outside the covered regions are then mined and turned into positive or negative prompts, while matched coverage centroids provide preservation prompts, enabling the shared SAM2 decoder to segment uncovered hard nuclei and reject stain-induced false prompts. Experiments on two public datasets, MoNuSeg and TNBC, show that StainPMS consistently improves over a strong auto-prompt baseline and achieves state-of-the-art performance.

Index Terms—Nuclei instance segmentation, histopathology, SAM2

I. INTRODUCTION

Nuclei instance segmentation in H&E histopathology is a fundamental step toward analyzing the tissue microenvironment. The count, morphology, and spatial distribution of individual nuclei underpin downstream tasks such as cancer grading and prognosis. Yet reliably localizing and delineating each nucleus in dense tissue is difficult, since instances frequently touch and overlap. This has driven a long line of dedicated segmentation methods.

Classical and deep nuclei-segmentation methods often reduce the problem to boundary separation or instance-center regression. Representative approaches predict auxiliary horizontal-vertical distance maps, direction fields, center heatmaps, or watershed-compatible representations to split touching nuclei [1]–[4]. More recently, segmentation foundation models such as SAM [5] and SAM2 [6] have been adapted to medical and histopathology images through medical adapters, auto-prompting, and prompt-free variants [7]–[14]. CA-SAM2 [12] follows this direction, combining a frozen SAM2 encoder, an automatic point head, and the SAM2 prompt encoder and mask decoder.

However, this formulation exposes a sharp bottleneck. Instances without prompts cannot be segmented, and difficult prompted instances may still have suboptimal mask agreement. Per-nucleus error analysis of the CA-SAM2 baseline shows that these residual failures are not random but concentrate on nuclei with weak contrast, small size, irregular morphology, or confusing context.

Our central observation is that the model’s residual errors are not entirely unsupported by the input image. Through color deconvolution, the hematoxylin channel emphasizes chromatin-rich nuclear material, so residual nuclei often remain visible as local hematoxylin peaks even when the model’s own prediction leaves them uncovered. This motivates StainPMS, a self-bootstrapped stain-guided Prompt-Mask Supervision framework (Fig. 1) that turns uncovered stain evidence into training-time supervision for the prompt-conditioned pathway, using no external coverage reference or separate teacher.

The main contributions are summarized below.

- A prompt-coverage deficiency is formulated for auto-prompted nuclei segmentation, namely that stain-visible nuclear evidence may remain underrepresented by the model’s current prompt-conditioned predictions. Building on this, StainPMS is introduced as a self-bootstrapped training framework that strengthens the existing CA-SAM2 prompt-mask pathway without adding learnable parameters or modifying inference.
- An online coverage map, periodically refreshed and monotonically accumulated from the model’s own predictions, guides hematoxylin residual mining. Stain peaks outside the dilated coverage become residual positive or negative prompts, while matched coverage centroids provide preservation prompts. Together these prompts supervise mask prediction.
- Extensive experiments on MoNuSeg and TNBC datasets demonstrate consistent improvements over previous methods such as the strong CA-SAM2 baseline.

II. PROPOSED METHOD

A. Framework Overview

StainPMS builds on the CA-SAM2 auto-prompt backbone, which comprises a frozen SAM2 large hierarchical vision

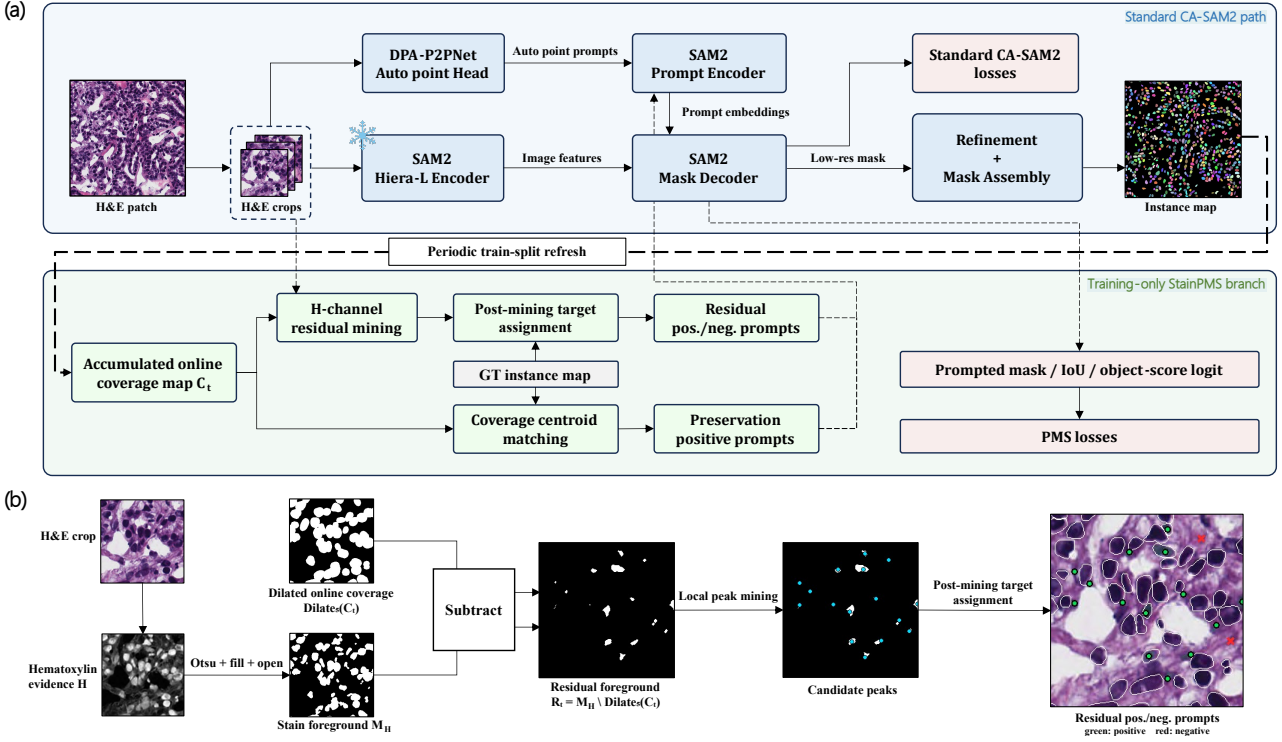


Fig. 1. Overview of the StainPMS framework. (a) The standard CA-SAM2 path is preserved, while a training-only PMS branch mines self-bootstrapped prompts and supervises the shared SAM2 mask decoder; only the CA-SAM2 path runs at inference. (b) In residual hematoxylin mining, the dilated online coverage map C_t is subtracted from the stain foreground M_H to expose residual foreground R_t , from which candidate prompts are drawn.

transformer (Hiera-L) encoder, a DPA-P2PNet [15] point head that proposes nuclear points, and the shared SAM2 prompt encoder and mask decoder that turn each point into an instance mask, a predicted Intersection-over-Union (IoU) score, and an object-score logit. On top of this pathway, StainPMS adds a training-only PMS branch that mines residual stain evidence outside the model’s own online coverage map C_t and supervises the shared decoder on the resulting prompts. Because the branch adds no parameters and is dropped at inference, the inference path and cost match the CA-SAM2 baseline. CA-SAM2 is chosen as the instantiation because it is the current accuracy leader, but the design requires only a point-prompted SAM-style decoder and can therefore strengthen any such method.

B. Self-Bootstrapped Online Coverage

The key contribution is that StainPMS constructs an online coverage map C_t during training, an instance-labeled map in which each covered nucleus carries a distinct integer ID, so individual covered instances can be distinguished instead of collapsing into a single foreground mask. Let A_θ denote the CA-SAM2 inference-and-assembly pipeline mapping a training image $x^{(n)}$ to an instance map. The initialized model generates the first coverage map $C_0^{(n)} = A_{\theta_0}(x^{(n)})$, and during training C_t is periodically refreshed by rerunning the current model on the training split, $\bar{C}_t^{(n)} = A_{\theta_t}(x^{(n)})$. The refresh is

not allowed to shrink the previous coverage; StainPMS applies a monotonic union operator,

$$C_t^{(n)} = C_{t-1}^{(n)} \cup_\tau \bar{C}_t^{(n)}, \quad (1)$$

where $\cup_\tau$ keeps every previous instance and admits a new predicted instance only when less than a fraction τ of its area is already covered; the added pixels are restricted to the previously uncovered region. This accumulation is essential, because without it a transiently weaker epoch would re-expose already-segmented nuclei as spurious residual failures, a destabilizing negative-feedback loop; monotonic growth instead makes the residual candidate set contract toward the genuinely hard regions. Because mining depends on the quality of C_t , PMS is activated only once the model yields a meaningful initial coverage map.

C. Stain-Guided Residual Mining

For each training crop, the RGB image is converted to hematoxylin-eosin-DAB (HED) space, and the hematoxylin channel obtained by color deconvolution [16], denoted H_0 , is used as the stain-evidence map. To suppress staining-intensity variation across slides, H_0 is robustly normalized using its first and ninety-ninth percentiles $q_1(\cdot)$ and $q_{99}(\cdot)$ and clipped to $[0, 1]$, namely,

$$\tilde{H} = \text{clip} \left(\frac{H_0 - q_1(H_0)}{q_{99}(H_0) - q_1(H_0)}, 0, 1 \right). \quad (2)$$

The normalized map is smoothed by a Gaussian filter to obtain H , and the stain foreground M_H is extracted using Otsu's method [17], hole filling, and morphological opening with a disk radius of 2.

To restrict mining to nuclei the model has not yet captured, the online coverage map C_t is binarized and dilated with a disk radius of 5, and the residual stain foreground is obtained by subtracting the covered region, namely,

$$R_t = M_H \setminus \text{Dilate}_5(\mathbf{1}[C_t > 0]). \quad (3)$$

Within R_t , local stain maxima are detected with a minimum spacing of 12 pixels and ranked by stain evidence, and the strongest are kept as the candidate set,

$$\mathcal{Q}_t = \text{TopK}_{20}\{p \in \mathcal{P}_{12}(H) : p \in R_t\}. \quad (4)$$

The dilation margin keeps chromatin variation inside a covered instance from being mistaken for a new nucleus.

D. Ground-Truth-Guided Prompt Assignment

The ground truth is used only to assign training labels to the mined prompts, never to mine candidates. For each candidate $c_i \in \mathcal{Q}_t$, StainPMS searches the ground-truth instance set $\mathcal{G}$ within radius $r = 8$ pixels,

$$y_i = \begin{cases} 1, & \exists G_j \in \mathcal{G} \text{ s.t. } d(c_i, G_j) \leq r, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

A positive candidate is paired with the matched instance mask, whereas an unmatched candidate is retained as a negative stain prompt for object-score rejection. To stabilize nuclei that are already covered by C_t , StainPMS also uses coverage-preservation prompts. For each instance in C_t , its centroid is matched to the ground truth within the same radius r , and matched centroids are added as positive prompts paired with their ground-truth masks. This anchors the pathway on already covered nuclei while the residual branch handles uncovered evidence.

E. Prompt-Mask Supervision

The assigned prompts are fed to the SAM2 prompt encoder during training. The image embedding and high-resolution features are shared with the main forward pass, and the shared SAM2 mask decoder is evaluated again on the stain-derived and preservation prompts.

The first term supervises the object-score logit, which decides whether a prompt corresponds to a real nucleus. Let $\mathcal{P}$ collect all PMS prompts in the crop, that is residual positives, residual negatives, and preservation positives. Binary cross-entropy (BCE) is applied to the object-score logit s_i of every prompt against its target y_i ($y_i = 1$ if the prompt is matched to a ground-truth instance, else 0),

$$\mathcal{L}_o = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \text{BCE}(s_i, y_i). \quad (6)$$

The remaining terms supervise the predicted masks and are applied only to positive prompts. For a positive prompt i , the

per-prompt mask losses are the focal, Dice, and IoU losses $\ell_f(i) = \mathcal{L}_{focal}(\hat{m}_i, m_i)$, $\ell_d(i) = \mathcal{L}_{dice}(\hat{m}_i, m_i)$, and $\ell_u(i) = \mathcal{L}_{iou}(\hat{u}_i, u_i)$, where $\hat{m}_i$ is the prompted mask, m_i the matched ground-truth instance mask, and $\hat{u}_i, u_i$ the predicted and target IoU. Averaging these losses over the residual positives $\mathcal{P}_{res}^+$ gives the residual mask loss

$$\mathcal{L}_k^{res} = \frac{1}{|\mathcal{P}_{res}^+|} \sum_{i \in \mathcal{P}_{res}^+} \ell_k(i), \quad k \in \{f, d, u\}, \quad (7)$$

which teaches the pathway to segment the newly mined nuclei. Averaging the same losses over the preservation positives $\mathcal{P}_{pre}^+$ gives the preservation mask loss

$$\mathcal{L}_k^{pre} = \frac{1}{|\mathcal{P}_{pre}^+|} \sum_{i \in \mathcal{P}_{pre}^+} \ell_k(i), \quad k \in \{f, d, u\}, \quad (8)$$

which keeps already-covered nuclei anchored as residual mining changes the prompt set.

These terms form the PMS loss. The object-score and residual terms use the global weight λ_{pms} ; residual mask losses are further scaled by λ_{res} , while preservation uses an independent weight λ_{pre} to anchor already-covered nuclei.

$$\begin{aligned} \mathcal{L}_{pms} = & \lambda_{pms} \mathcal{L}_o + \lambda_{pms} \lambda_{res} (\mathcal{L}_f^{res} + \mathcal{L}_d^{res} + \mathcal{L}_u^{res}) \\ & + \lambda_{pre} (\mathcal{L}_f^{pre} + \mathcal{L}_d^{pre} + \mathcal{L}_u^{pre}). \end{aligned} \quad (9)$$

The final training objective is

$$\mathcal{L} = \mathcal{L}_{det} + \mathcal{L}_{mask} + \mathcal{L}_{pms}. \quad (10)$$

Here $\mathcal{L}$ denotes the final training objective, and $\mathcal{L}_{det}$ and $\mathcal{L}_{mask}$ denote the standard point-detection and mask-supervision losses of the CA-SAM2 backbone, respectively.

III. EXPERIMENTS

A. Datasets and Metrics

StainPMS is evaluated on MoNuSeg [18] and TNBC [19], two H&E nuclei instance-segmentation benchmarks. MoNuSeg consists of 1000×1000 -pixel images spanning seven organs, whereas TNBC contains 512×512 -pixel images from eleven triple-negative breast cancer patients. In our experiments, MoNuSeg is split into 37 training and 14 testing images, and TNBC is split into 37 training and 13 testing images. Performance is reported using Dice, aggregated Jaccard index (AJI), detection quality (DQ), segmentation quality (SQ), and panoptic quality (PQ) [20]. PQ is computed as $PQ = DQ \times SQ$ under the hard IoU ≥ 0.5 matching threshold.

B. Training and Evaluation Protocol

StainPMS supports both warm-started fine-tuning and from-scratch training. With a well-performing pretrained initialization PMS is enabled immediately, since the initial model already gives a meaningful coverage map; without one, the model first runs the standard objective for $K = 50$ warm-up epochs before PMS is activated. In the latter setting, training uses a cosine learning rate from 1×10^{-4} to 1×10^{-6} , refreshes

TABLE I
COMPARISON WITH PRIOR METHODS ON MoNuSeg AND TNBC.

Dataset	Model	Prompt	Dice	AJI	DQ	SQ	PQ
MoNuSeg	U-Net [21]	×	0.801	0.573	0.666	0.739	0.495
	Mask R-CNN [22]	×	0.760	0.546	0.521	0.700	0.374
	HoVer-Net [1]	×	0.803	0.595	0.743	0.754	0.563
	SMILE [2]	×	0.782	0.576	0.709	0.749	0.533
	PointNu-Net [3]	×	0.792	0.582	0.754	0.750	0.568
	SAM (Zero-Shot) [5]	×	0.448	0.188	0.106	0.562	0.069
	MedSA [8]	×	0.669	0.361	0.359	0.671	0.242
	HQ-SAM [23]	×	0.663	0.526	0.685	0.759	0.519
	CellViT [24]	×	0.797	0.584	0.722	0.749	0.543
	PromptNucSeg-L [25]	Auto-point	0.804	0.607	0.767	0.758	0.584
	SAM2 [6]	One-point	0.732	0.382	0.377	0.673	0.257
	MedSAM2 [11]	One-point	0.735	0.394	0.403	0.675	0.276
	CA-SAM2 [12]	Auto-point	0.808	0.644	0.826	0.750	0.620
	StainPMS	Auto-point	0.822	0.666	0.853	0.771	0.658
TNBC	CA-SAM2	Auto-point	0.787	0.639	0.835	0.808	0.676
	StainPMS	Auto-point	0.808	0.665	0.838	0.813	0.682

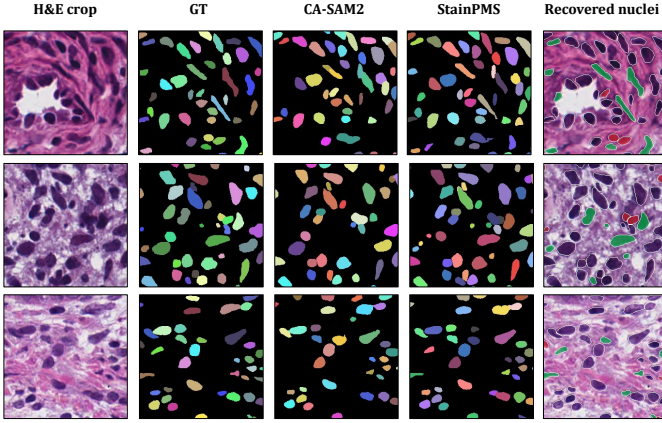


Fig. 2. Qualitative comparison on MoNuSeg. Columns show the input crop, ground truth, CA-SAM2, StainPMS, and the correction overlay. In the overlay, green marks nuclei whose IoU crosses 0.5 only after StainPMS, red marks nuclei below 0.5 for both methods, and white contours are ground-truth boundaries.

the coverage map every $T = 20$ epochs, and weights PMS with $\lambda_{pms} = 0.5$, $\lambda_{res} = 0.3$, $\lambda_{pre} = 1.0$, at most 20 preservation prompts per crop, and a monotonic-union threshold $\tau = 0.5$. Both datasets use 256×256 crops, batch size 1, and AdamW with weight decay 10^{-4} on a single NVIDIA RTX 4090 GPU; the tiling overlap is 32 pixels for TNBC and 92 pixels for MoNuSeg. Results are reported at non-maximum suppression (NMS) threshold 12 unless otherwise specified, with other thresholds used only for robustness analysis, and no test-time augmentation.

C. Comparison with Existing Methods

Table I compares StainPMS with classical segmenters, prompt-based foundation models, and the CA-SAM2 baseline; prior results use their published settings, and the controlled comparison is CA-SAM2 versus StainPMS under the same inference path. On MoNuSeg, StainPMS improves all five metrics together, adding 2.2% AJI and 3.8% PQ over CA-SAM2 and 5.9% AJI and 7.4% PQ over the strongest prior segmenter PromptNucSeg-L. On TNBC the gain concentrates in AJI (2.6%) while DQ stays flat, reflecting an IoU-cliff

TABLE II
PQ UNDER FOUR NMS THRESHOLDS.

Dataset	Method	thr=2	thr=6	thr=12	thr=18
MoNuSeg	CA-SAM2	0.631	0.627	0.620	0.620
MoNuSeg	StainPMS	0.663	0.659	0.658	0.655
TNBC	CA-SAM2	0.679	0.678	0.676	0.676
TNBC	StainPMS	0.680	0.680	0.682	0.680

TABLE III
ABLATION STUDY ON THE COMPONENTS OF STAINPMS ON MoNuSeg.

Variant	AJI	SQ	PQ
CA-SAM2 baseline	0.644	0.750	0.620
StainPMS w/o object-score	0.662	0.766	0.651
StainPMS w/o coverage accumulation	0.656	0.766	0.645
StainPMS w/o preservation prompts	0.665	0.762	0.651
StainPMS full	0.666	0.771	0.658

effect. Since PQ matches only at $\text{IoU} \geq 0.5$, masks whose IoU rises but stays below 0.5 lift AJI without adding matched detections. MoNuSeg nuclei more often cross this threshold and convert into true positives (green in Fig. 2).

D. Robustness to the NMS Threshold

Table II reports PQ under four NMS thresholds. StainPMS stays above the baseline across all eight dataset-threshold combinations. MoNuSeg favors smaller thresholds while TNBC is nearly insensitive, so the gain reflects training-time supervision rather than a particular NMS setting; main results use threshold 12.

E. Ablation Study

Table III isolates each component on MoNuSeg. Removing any one lowers PQ yet stays above the CA-SAM2 baseline, so the mechanisms are complementary. Monotonic coverage accumulation matters most, with the largest AJI and PQ drop when removed, consistent with keeping covered nuclei out of the residual pool as predictions fluctuate between refreshes. Preservation prompts mainly protect segmentation quality, producing the largest SQ drop when removed while leaving AJI almost unchanged, and object-score rejection has a smaller, broader effect on both detection and mask quality.

IV. CONCLUSION

StainPMS is a self-bootstrapped stain-guided framework for prompt-conditioned nuclei instance segmentation. It mines residual hematoxylin peaks outside the model's own accumulated coverage and supervises the resulting prompts, so the shared pathway learns from stain evidence that signals omissions or hard regions while covered nuclei stay anchored. Experiments on MoNuSeg and TNBC show state-of-the-art performance. The benefit is conditional on residual stain evidence containing informative hard nuclei. Because the supervision acts only at training time and leaves the inference path untouched, it can in principle be attached to other prompt-conditioned backbones; future work may extend it to additional stain channels and to multi-class nuclei classification.

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